SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA5117 - Cryptography and Network Security

ASSIGNMENT 1

QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.
- Detect tampering with prescriptions, dosage and lab values.
- Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge : An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between

hospitals. Re-evaluate your design.

- Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.
- Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.

- Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.
- Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.
- Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it. Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.
- Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks **4 Marks**

Identify at least six security attacks relevant to the healthcare scenario.

- Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).
- Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.
- Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.
- Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

TASK 2 – Map and Judge Security Goals **3 Marks**

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

- Judge which goals classical ciphers can and cannot deliver.
- Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

TASK 3 – Apply and Critique the Bell-LaPadula Model **5 Marks**

- Define the security levels and subject/object assignments for the network.
- Apply the simple security property (no read up) and the *-property (no write down) to at least five access requests, including two that must be denied. Show the reasoning for each.

- Evaluate the model critically in a medical setting: what happens in an emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.
- Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether Bell-LaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

TASK 4 – Design the Cryptographic Strategies **4 Marks**

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

- The encryption and decryption algorithm (pseudocode or formula) and the key definition.
- A worked example on a 20+ character message from the scenario.
- A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

TASK 5 – Mathematical Analysis of the Schemes **5 Marks**

- Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations).
- For Hill: show the conditions under which K^{-1} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.
- Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).
- Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

TASK 6 – Evaluate Cryptanalytic Resistance **4 Marks**

Attack each chosen scheme using a technique suited to it:

- Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.
- Explain why each attack succeeds or fails, in terms of letter-frequency preservation, key repetition and linearity.
- Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

TASK 7 – Assess Transposition and Product Ciphers **3 Marks**

- Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.
- Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.
- Judge whether the added complexity is practical for rural clinic staff doing manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

TASK 8 – Evaluate Steganography **3 Marks**

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

- Compare it with encryption on capacity, imperceptibility, robustness and detectability.
- Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.
- Judge when steganography is appropriate, and argue why hiding is not a substitute for encrypting. Evaluate an encrypt-then-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

TASK 9 – Implement and Test **4 Marks**

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

- Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).
- Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.
- State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure

TASK 10 – Record and Interpret Experimental Evidence **3 Marks**

Present data in tables and graphs for:

- Letter-frequency histograms of plaintext versus each ciphertext.
- Index of coincidence per scheme.
- Encryption/decryption time against message size.
- Attack success rate against ciphertext length.

Evaluate what the data shows. Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

- Justify against the alternatives using your mathematical and experimental evidence.
- State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.
- Address key distribution across 25 clinics and usability for non-technical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid limitations.

DAY- 4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

- What has changed in the threat model (insider versus external attacker)?
- Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?
- Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks)

- Test your recommended scheme against the new attacker capabilities (known-plaintext, repeated-key attacks).
- Support the evaluation with a demonstration or calculation.
- Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

- Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.
- Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and key-management cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

MARK DISTRIBUTION

Task	Marks
Task 1 – Threats, Attacks and Risk Evaluation	4
Task 2 – Security Goals Mapping	3
Task 3 – Bell-LaPadula Application and Critique	5
Task 4 – Cipher Design	4
Task 5 – Mathematical Analysis	5
Task 6 – Cryptanalytic Resistance	4
Task 7 – Transposition / Product Ciphers	3
Task 8 – Steganography Evaluation	3
Task 9 – Implementation & Testing	4
Task 10 – Experimental Evidence	3
Task 11 – Final Design Recommendation	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	**8**
TOTAL	**50**

SIMATS ENGINEERING

CSA5117 – Cryptography and Network Security

ASSIGNMENT 1 – CO1

Design, Analyse and Validate Classical Encryption Techniques for Securing Sensitive Information

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
		CSE	2026–2027	Dr. D. Sudhagar	Individual	1 Week	50		

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
1	Cipher Design and Selection	5		Select and describe three suitable classical ciphers (Caesar, monoalphabetic substitution, and Vigenère or Hill); justify their use and limitations for the given sensitive message.
2	Encryption and Decryption	5		Show correct key selection, encryption steps, ciphertext, decryption steps and recovery of the original plaintext.
3	Key Analysis and Mathematical Basis	5		Explain key space, modular arithmetic or alphabet mapping used by each selected cipher; calculations must be traceable.
4	Security and Cryptanalysis	5		Analyse vulnerabilities such as brute-force search, frequency analysis, repeated-key patterns and known-plaintext attacks where relevant.
5	Comparison of Cipher Strength	4		Compare the selected methods using key space, implementation effort, likely attacks and suitability for sensitive information.
6	Algorithm / Pseudocode Formulation	4		Provide clear encryption and decryption pseudocode or flowcharts, including input validation and treatment of spaces/punctuation.
7	Worked Example and Verification	4		Use a sample message; verify that decrypt(encrypt(plaintext)) reproduces the original message under the stated rules.
8	Implementation and Testing	4		Implement at least two ciphers in a programming language; include source code, test cases, expected output and actual output.
9	Experimental Evaluation	3		Test multiple messages and key choices; present organised tables of ciphertext, recovered plaintext, execution time or operation counts.
10	Theory-to-Result Discussion	3		Explain how observed results match the cipher design and why classical ciphers are not suitable for modern high-security use.
11	Recommend and Defend a Design	4		Recommend a method for the assignment scenario and defend the decision using correctness, usability and security limitations.
12A	Re-evaluate: Impact Analysis	2		Discuss the impact of message length, key choice and repeated patterns on confidentiality.
12B	Re-evaluate: Attack Evaluation	2		Evaluate likely attacks against the selected ciphers and identify information an attacker could exploit.
12C	Re-evaluate: Adaptation or Redesign	2		Propose a safer modern replacement or layered approach, explaining why classical ciphers are only educational examples.
12D	Re-evaluate: Evidence-Based Defence	2		Defend conclusions using worked examples, test results, key-space reasoning and a comparison table.
TOTAL		50		Marks to be awarded by the faculty.

OVERALL REMARKS

Faculty feedback: _____

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Student Signature	Faculty Signature	Program Director
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MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present X = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
1	Cipher selection, design rationale and assumptions	☐	Three ciphers described and justified
2	Encryption and decryption working shown	☐	Steps, key and plaintext/ciphertext included
3	Mathematical basis and key-space explanation	☐	Calculations are traceable
4	Security analysis and likely attacks	☐	Relevant weaknesses identified
5	Cipher comparison table	☐	Criteria applied consistently
6	Pseudocode or flowchart	☐	Encryption and decryption steps included
7	Worked example and correctness verification	☐	Recovered plaintext matches original
8	Source code and reproducible tests	☐	Program, test inputs and outputs included
9	Experimental results in tables	☐	Multiple inputs or keys evaluated
10	Theory and experimental results compared	☐	Observations interpreted
11	Recommendation supported by analysis	☐	Trade-offs and limitations stated
12A–D	Impact, attack evaluation, redesign and evidence-based defence	☐	All four parts addressed

IMPLEMENTATION EVIDENCE CHAIN (Tasks 8–9)

Evidence Item	Status	Observation
Source code	☐	Readable source code for each implemented cipher; language and execution instructions stated
Test inputs	☐	Plaintexts, keys and edge cases recorded
Expected behaviour	☐	Expected ciphertext and recovered plaintext documented
Actual output	☐	Program output or screenshots included
Intermediate evidence	☐	Key mapping, modular steps or intermediate transformations shown
Boundary / failure tests	☐	Empty input, spaces, punctuation, invalid keys and repeated characters considered
Interpretation	☐	Results connected to cipher properties and security limitations
Code, data and report consistent	☐	Tables and claims match the submitted program output

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All assignment tasks including 12A–12D answered	☐ Yes ☐ Partly ☐ No
Encryption and decryption shown for each selected cipher	☐ Yes ☐ Partly ☐ No
Keys and assumptions clearly stated	☐ Yes ☐ Partly ☐ No
At least two ciphers implemented and tested	☐ Yes ☐ Partly ☐ No
Same plaintext conventions used consistently	☐ Yes ☐ Partly ☐ No
Source code and reproducible test procedure provided	☐ Yes ☐ Partly ☐ No
Tables / figures / program outputs included	☐ Yes ☐ Partly ☐ No
Cross-references and numbering correct	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders in the student report	☐ Yes ☐ Partly ☐ No
Viva recommended where implementation or calculations need clarification	☐ Yes ☐ No

ANSWERS – ASSIGNMENT 1

CSA5117 – Cryptography and Network Security | CO1

Note: Answers are concise and follow the marks allocated in the question paper. Classical cipher examples use A=0, B=1, …, Z=25. Spaces and punctuation are retained only where stated.

TASK 1 – Evaluate Threats, Attacks and Risks (4 Marks)

Threat is a possible cause of harm; vulnerability is a weakness; attack is an attempt to exploit a weakness; risk is the likelihood and impact of resulting harm. Example: a dishonest clinic employee is a threat, a shared key is a vulnerability, copying ciphertext is an attack, and exposure of patient diagnoses is the risk.

Attack	Type	Likelihood (1–5)	Impact (1–5)	Risk score	Reason
Prescription/dosage modification	Active	4	5	20	May cause incorrect treatment or death.
Replay of an old prescription	Active	4	5	20	An obsolete order may be accepted as current.
Eavesdropping	Passive	4	4	16	Exposes diagnoses and personal information.
Masquerade as a doctor	Active	3	5	15	Unauthorised clinical instructions may be trusted.
Traffic analysis	Passive	4	3	12	Reveals patient activity and clinic relationships.
Denial of service	Active	3	4	12	Delays access to records and treatment.

The first priority is prescription modification, followed closely by replay, because both can directly harm patients. Integrity checks, unique order numbers, timestamps and clinical confirmation are essential.

TASK 2 – Map and Judge Security Goals (3 Marks)

System function	Most important goals	Reason / limitation
Record upload	Confidentiality, integrity, authentication, availability	Records must remain private, complete and accessible to authorised clinics.
Prescription transmission	Integrity, authentication, confidentiality, non-repudiation	The dosage and prescriber must be verified; old orders must be rejected.
Specialist consultation	Confidentiality, authentication, availability, integrity	Only the intended care team should read and rely on the advice.

Classical ciphers mainly conceal message contents. They do not reliably provide integrity, authentication or non-repudiation. Successful decryption only shows that a key was used; it does not prove who sent the message or that its dosage was not altered.

TASK 3 – Apply and Critique the Bell–LaPadula Model (5 Marks)

Security levels, from highest to lowest, are Top Secret, Secret, Confidential, Restricted and Public. Assign the Medical Director to Top Secret, specialist doctors to Secret, nurses and laboratory staff to Confidential, billing/front-desk staff to Restricted, and patients to Public. Compartments such as oncology can further restrict access.

Subject	Requested object	Decision	Reason
Medical Director (Top Secret)	Secret specialist report	Allow read	No read up is satisfied.
Specialist (Secret)	Confidential lab result	Allow read	Reading a lower level is permitted.
Nurse (Confidential)	Secret allergy record	Deny read	Violates no read up.
Billing staff (Restricted)	Confidential diagnosis	Deny read	Violates no read up.
Specialist (Secret)	Restricted billing file	Deny write	Violates no write down.
Nurse (Confidential)	Restricted record	Deny write	Violates no write down.

The simple-security property means no read up; the *-property means no write down. In an emergency, a nurse needing a Secret allergy record may use a controlled break-glass process with temporary, logged, purpose-limited access. A doctor writing a billing note should use an authorised trusted subject or a reviewed declassification process that releases only necessary information.

Bell–LaPadula protects confidentiality, but not integrity. It may be rigid during emergencies and does not eliminate covert channels. It is useful as one access-control layer, but is not sufficient alone for healthcare.

TASK 4 – Design the Cryptographic Strategies (4 Marks)

1. Caesar cipher

Key: k = 3. Encryption: C = (P + k) mod 26. Decryption: P = (C − k) mod 26. Example plaintext: PATIENTRECORD. Ciphertext: SDWLHQWUHFRUG. Shifting each letter back by three recovers PATIENTRECORD. It is simple but has only 25 non-trivial keys.

2. Vigenère cipher

Key: MED, repeated. Encryption: $C_i = (P_i + K_i) \bmod 26$. Decryption: $P_i = (C_i - K_i) \bmod 26$. For plaintext PATIENTRECORD and key MEDMEDMEDMED, the ciphertext is BEWUIQFVHOSU. Reversing the modular addition recovers the plaintext. It hides simple single-letter patterns better than Caesar, but repeated keys can be attacked.

3. Columnar transposition

Write PATIENTRECORD row-wise into four columns, pad the last row with X if needed, and read columns in the agreed key order (for example 3–1–4–2). Decryption restores the columns and reads row-wise. This rearranges letter positions but preserves the letter-frequency counts, so it provides limited resistance to statistical analysis.

These schemes are meaningfully different: Caesar and Vigenère substitute letters, while columnar transposition rearranges positions.

TASK 5 – Mathematical Analysis of the Schemes (5 Marks)

Caesar: 25 non-trivial shifts (26 including the identity). Monoalphabetic substitution: $26! \approx 4.03 \times 10^{26}$ keys. Vigenère with a fixed key length m: 26^m keys. Columnar transposition with n columns: n! column orders. Hill cipher: valid keys are invertible matrices modulo 26.

For a Hill key K = [[3, 3], [2, 5]], det(K) = 3×5 − 3×2 = 9. Since gcd(9,26) = 1, the key is invertible. The inverse of 9 modulo 26 is 3. Therefore $K^{-1} = 3 \times [[5,-3],[-2,3]] \bmod 26 = [[15,17],[20,9]]$. Multiplication gives the identity matrix modulo 26.

A large key space does not guarantee security. Monoalphabetic substitution has 26! keys but preserves letter-frequency patterns and common digrams, so frequency analysis can recover the mapping.

TASK 6 – Evaluate Cryptanalytic Resistance (4 Marks)

Scheme	Suitable attack	Why it works
Caesar	Brute force and frequency analysis	Only 25 non-trivial shifts; all can be tested.
Monoalphabetic	Frequency and digram analysis	Each plaintext letter always maps to the same ciphertext letter.
Vigenère	Kasiski examination / index of coincidence	A repeated key creates periodic patterns.
Hill	Known-plaintext attack	Linear equations can reveal the key from enough independent blocks.
Columnar transposition	Anagram and digram analysis	Letter counts remain unchanged.

For these basic implementations, Vigenère with a sufficiently long key generally resists simple frequency analysis best, followed by Hill against ciphertext-only analysis, columnar transposition, monoalphabetic substitution and Caesar. Repeated-key Vigenère and Hill with known plaintext can still be broken.

TASK 7 – Assess Transposition and Product Ciphers (3 Marks)

In columnar transposition, plaintext is written in rows and read in a secret column order. It disrupts the positions of adjacent letters but preserves the exact letter counts. Substitution followed by transposition is stronger than either alone because it changes letter identities and rearranges positions. However, frequency information remains and longer ciphertext can support attacks. Manual use is inconvenient for rural clinics, so software automation would be necessary.

TASK 8 – Evaluate Steganography (3 Marks)

In least-significant-bit (LSB) embedding, convert a short report into bits and replace selected image pixels' least significant bits with those message bits. Extraction reads the same positions in order. Capacity is about one bit per selected pixel for a one-bit method. The visual change may be small, but statistical tests such as chi-square can detect altered distributions; JPEG compression can destroy the hidden message.

Criterion	Encryption	LSB steganography
Purpose	Makes content unreadable without a key	Conceals the existence of a message
Capacity	Depends on message and cipher format	Limited by available image samples
Robustness	Ciphertext remains usable if copied intact	Fragile under JPEG compression or resizing

Detectability	Encrypted data may be recognisable	May be statistically detectable
Medical-image risk	Does not intentionally alter diagnostic pixels	Pixel changes may affect diagnosis

Diagnostic pixels should not be modified. If hiding is required, use a separate lossless carrier or a non-diagnostic region and preserve the original image. Encrypt the report before embedding it. Steganography is not a substitute for encryption.

TASK 9 – Implement and Test (4 Marks)

The following Python code implements Caesar and Vigenère. It retains spaces, digits and punctuation unchanged, and verifies that decryption recovers the original message.

```
import string
A = string.ascii_uppercase

def caesar(text, k):
    out = ""
    for ch in text:
        if ch.upper() in A:
            i = A.index(ch.upper())
            x = A[(i + k) % 26]
            out += x if ch.isupper() else x.lower()
        else:
            out += ch
    return out

def vigenere(text, key, decrypt=False):
    key = key.upper()
    if not key or any(c not in A for c in key):
        raise ValueError("Key must contain A-Z only")
    out, j = "", 0
    for ch in text:
        if ch.upper() in A:
            shift = A.index(key[j % len(key)])
            if decrypt:
                shift = -shift
            x = A[(A.index(ch.upper()) + shift) % 26]
            out += x if ch.isupper() else x.lower()
            j += 1
        else:
            out += ch
    return out

message = "PATIENT ID PT-20481; AMOX 500MG."
for encrypt, decrypt in [
    (lambda x: caesar(x, 3), lambda x: caesar(x, -3)),
    (lambda x: vigenere(x, "MED"), lambda x: vigenere(x, "MED", True))
]:
    cipher = encrypt(message)
    recovered = decrypt(cipher)
    assert recovered == message
    print("Ciphertext:", cipher)
    print("Recovered:", recovered)
```

Testing procedure: run in Python 3; confirm the recovered text exactly matches the input; test empty text, punctuation, lowercase letters, digits and invalid keys. For performance comparison, use identical messages of 100, 1,000, 10,000 and 100,000 characters and record actual encryption/decryption times. Do not report invented timings.

TASK 10 – Record and Interpret Experimental Evidence (3 Marks)

Measurement	Expected observation	Interpretation
Letter-frequency histogram	Caesar and monoalphabetic show permuted frequencies; transposition retains exact counts; Vigenère tends to flatten frequencies.	Frequency preservation enables classical attacks.
Index of coincidence	Caesar, monoalphabetic and transposition remain language-like; Vigenère usually lowers IC when the key is suitable.	Lower IC may indicate polyalphabetic encryption.
Execution time	Simple algorithms generally scale with message length.	Record measured time on the same computer.
Attack success	Caesar is quickly broken; repeating-key Vigenère becomes vulnerable as ciphertext grows.	More ciphertext may help the attacker.

Present histograms, IC values and measured timings in tables or graphs. The actual numerical results must come from running the implementation. Interpret them using the weaknesses explained in Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

For an educational classical-only design, use separate per-clinic Vigenère keys, rotate them regularly, and apply Bell–LaPadula access labels and compartment checks. Avoid one network-wide key. Log access and require controlled approval for emergency access. Classical ciphers remain vulnerable to known-plaintext attacks, insider key theft, modification and impersonation.

For real patient records, use modern authenticated encryption, digital signatures or equivalent identity controls, secure key management, audit logs, backups and availability controls. Classical methods are suitable for learning, not real patient-data protection.

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

A. Impact Analysis (2 Marks)

The threat now includes an insider who has copied many ciphertexts. The assumptions that every clinic keeps a shared key secret, that keys are not reused and that message formats are unpredictable are no longer valid. Predictable headers provide known plaintext and repeated messages reveal more key information.

B. Evaluation of the Original Design (2 Marks)

In a repeating-key Vigenère cipher, if plaintext and ciphertext are known, key letters are calculated by $K_i = (C_i - P_i) \bmod 26$. A predictable header such as PATIENT ID therefore exposes corresponding key positions. Repeated messages expose the same positions again. Bell–LaPadula still controls labelled access, but cannot stop an authorised insider who already possesses the key from copying or decrypting data.

C. Adaptation or Redesign (2 Marks)

Modify the educational design by assigning a distinct key to each clinic and a fresh session key for each exchange; rotate and revoke keys; reduce predictable headers; add random padding; restrict key access; and log key use. Encrypt a report before embedding it in a lossless image carrier. Never alter diagnostic pixels. These measures reduce reuse but do not make classical ciphers secure by modern standards.

D. Evidence-Based Defence (2 Marks)

A Vigenère key of length m has 26^m possible strings, compared with 25 non-trivial Caesar shifts. Yet known plaintext can reveal key positions, so key-space size alone is not proof of security. More keys also increase distribution and rotation workload across 25 clinics. A real network needs managed key distribution and modern authenticated cryptography. Classical techniques cannot guarantee modern confidentiality, integrity, authentication or non-repudiation.

CONCLUSION

Classical ciphers demonstrate substitution, transposition, modular arithmetic and access-control reasoning. They remain vulnerable to statistical and known-plaintext attacks, especially when keys are reused and message formats are predictable. Bell–LaPadula helps protect confidentiality but does not ensure integrity or provide sufficient emergency flexibility. The classical design is appropriate for

academic analysis only; real healthcare systems require modern authenticated encryption, secure key management, identity verification and audited emergency access.